

ABDUL RAFAY KHAN

SOFTWARE DEVELOPER

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PROFILE SUMMARY

Motivated final-year Computer Science student with a solid foundation in programming fundamentals, object-oriented design, and full-stack web development (ASP.NET Core Web API, Entity Framework Core, RESTful APIs, MVC architecture). Strong written and verbal communicator. Experienced with Git/GitHub, JWT authentication, and database-backed CRUD applications (PostgreSQL, MySQL, Firebase). Eager to deepen expertise in a Python (Django/FastAPI) or MERN track through a hands-on, and to grow as a professional software developer through code reviews, agile teamwork, and real-world projects.

SKILLS

Programming Languages: Java, C++, C#, Python, JavaScript (ES6 basics), PHP

Web & Backend: ASP.NET Core Web API/MVC, Entity Framework Core, RESTful APIs, JWT Auth, HTML5, CSS3, Bootstrap, Next.js

Database: PostgreSQL, MySQL, Firebase Realtime Database, SQLite

Tools & Practices: Git, GitHub, VS Code, Android Studio, Agile/Sprint basics

Currently Learning: Django/FastAPI, React, Node.js/Express, Flutter

ACADEMIC PROJECTS

InterviewAce – AI Interview Preparation Platform

AI-driven platform for personalized interview practice, resume analysis, AI-generated questions, and performance insights. Built a scalable backend with ASP.NET Core Web API, Entity Framework Core, PostgreSQL, JWT authentication, and RESTful API design.

Multiclass Land Cover Recognition – Explainable Deep Learning Framework

Designed a deep learning framework using Convolutional Neural Networks (CNNs) to classify Land Use and Land Cover (LULC) types from remote sensing imagery, addressing complex spatial-spectral variation across diverse environmental conditions. Integrated Explainable AI (XAI) methods — including Score-CAM, LIME, SHAP, Saliency Maps, Integrated Gradients, SmoothGrad, Gradient x Input, Occlusion, and DeepLIFT — to make model predictions transparent and interpretable for decision-makers. Applied preprocessing pipelines and transfer learning to improve model generalization across varied geospatial datasets. Validated using 5-fold cross-validation (mean accuracy $98.40\% \pm 0.06\%$, Macro-F1 98.38%, MCC 0.984) and achieved 99.65% overall accuracy with a final ensemble model, alongside strong calibration and a low failure rate (0.35%).

Task Manager – .NET Core MVC Web App

Task management system built with ASP.NET Core MVC (3-layer architecture), RESTful API endpoints, and MySQL for full CRUD operations.

AV Messenger – Real-Time Chat App

Real-time messaging app in Java/Android using Firebase Realtime Database for live sync and Firebase Authentication for secure login.

EDUCATION

BS Computer Science

2023 – 2027

National University of Modern Languages (NUML)

ICS Physics

2021 – 2022

Punjab Group of Colleges

Matriculation

2019 – 2020

M.N Education Campus School

ADDITIONAL INFORMATION

Languages: English (Proficient), Urdu (Native), French (Basic)

Interests: Web Development, Applied Machine Learning, Open-Source Contribution, Learning Python & React